

Zhongshu Xu

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EDUCATION

The Ohio State University

Aug 2021 - May 2026

Ph.D. in Applied Mathematics

- **Relevant courses:** Probability, Machine Learning, Intermediate Studies in Artificial Intelligence

University of Science and Technology of China

Aug 2016 - Jun 2020

B.S. in Information and Computing Sciences (Applied Mathematics)

- **Relevant courses:** C, Java, Data Structure and Algorithm, Mathematical Statistics

INDUSTRY EXPERIENCE

Quantitative Analysis Intern (Model Risk Governance and Review) | JPMorgan Chase & Co.

Jun 2025 – Aug 2025

- Conducted local and global explainability analysis for Check Image Model with **YOLO** and **ResNet50** backbones, applying black-box (RISE, LIME, SINEX) and white-box (Grad-CAM, LRP) techniques to generate saliency heatmaps and identify key handwriting features influencing model decisions.
- Identified redundancy via global embedding-space analysis, enabling removal of the last two ResNet50 layers, achieving a 94% reduction in parameters and a 5x speedup with no loss in accuracy.

Machine Learning Intern | Riverain Technologies

May 2024 – Aug 2024

- Developed a deep learning pipeline for artery/vein segmentation in pulmonary vessel CT scans, combining a **UNet**-based centerline extraction model with a graph-based separation algorithm, achieving a Dice score above 0.98.
- Enhanced model robustness to real-world artifacts—such as missing data, discontinuities, and overlapping masks—through targeted output space design and synthetic data augmentation.

Machine Learning Intern | Ask Mendel AI

Aug 2023 – Dec 2023

- Designed a core strategy for **LLM**-based code generation using **Langchain**, integrating Knowledge Graphs, Vector Stores, and Multi-Agent coordination to improve the efficiency and accuracy of data analysis and visualization.
- Collaborated with bioinformatics experts to build a domain-specific dataset for fine-tuning Llama 2 / Code Llama, boosting model accuracy and performance by 25% on specialized biomedical tasks.

Quantitative Analyst Intern | Zhejiang Yingyang Asset Management Co., Ltd.

Sep 2020 – Jan 2021

- Devised and implemented a stock prediction system for the CSI-500 index using **RNN**, **LSTM**, and **Transformers**, improving prediction accuracy by 15%.
- Led a three-person team in integrating multiple forecasting models, resulting in a trading strategy that is more resilient and robust. Our model achieved higher performance in backtesting compared to the company's average.

RESEARCH PROJECTS

Digital Twin for Airfoil Optimization with Reinforcement Learning

Feb 2024 – Present

- Developed a **Transformer** model to forecast airfoil surface pressure distributions from historical data, achieving inference speeds **100× faster** than traditional CFD simulations.
- Embedded the model into **Deep Q-Learning** framework as surrogate environment, enabling the agent to dynamically control the airfoil's angle of attack to maximize the lift-to-drag ratio and accelerate aerodynamic optimization.

Data-Driven Modeling of Differential Equations (SDEs & PDEs)

Aug 2021 – Present

- Developed **Density-Enforced Auto-Encoders** to model underlying **Stochastic Differential Equations (SDEs)** from time-series data, reducing prediction error by **5x** compared to traditional methods like VAE and GAN. Paper published in [JMLMC](#).
- Created a memory-based neural network structure, designed specifically for PDE modeling in scenarios where data is incomplete, expanded the potential applications of neural networks in PDE modeling. Paper published in [JCP](#).
- Engineered an efficient **spatio-temporal Transformer model** for forecasting **Partial Differential Equations (PDEs)** solutions, significantly improving long-term accuracy with only **10% of parameters** required by standard video Transformers.

PROFESSIONAL SKILLS

Programming Languages: Python (TensorFlow, PyTorch, Langchain, Pandas), R, SQL, C/C++, MATLAB, Mathematica

Platforms & Tools: Linux, Gitlab, Docker, Amazon SageMaker, AWS S3